

ISHAAN MALHOTRA

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Technical Skills

Software: Python, ROS2, MATLAB, Simulink, RViz, Solidworks, Unreal Engine, Unity, Git

Libraries & Frameworks: PyTorch, sci-kit learn, OpenCV, NumPy, pandas, Matplotlib

Specializations: Autonomous Vehicles, Deep Learning, Electric Vehicles, Reinforcement Learning, MLOps

Experience

Distributed Robotics and Networked Embedded Systems Lab- MOOG

Buffalo, NY

Graduate Research Assistant (P.I: Dr. Karthik Dantu)

Jan 2024 -Present

- Collaborating with MOOG Construction on developing an autonomous excavator, retrofitted with Nvidia Orin Jetsons running ROS2 for autonomous control.
- Integrated GPS, IMUs, LiDARs, and stereo cameras for enhanced perception and autonomous operation.
- Performed visual-inertial calibration using ArUco markers for precise intrinsic and extrinsic alignment.
- Designed and built a stereo camera system using two monocular telephoto cameras with a high baseline to enhance depth perception and accuracy in challenging environments.

Danfoss Power Solutions

Ames, IA

Autonomous Simulation Engineer Intern

May 2024 -August 2024

- Simulated autonomous software for LiDAR-based off-highway robots using Unreal Engine, Simulink, and ROS.
- Implemented a structured data collection experiment to validate simulation results for an autonomous robot.
- Developed an LSTM-based pipeline for reduced-order black-box modeling of the vehicle plant.
- Leveraged Functional Mock-up Units (FMUs) to seamlessly integrate and simulate C++ code within Simulink.

Daimler Truck Innovation Center India (Mercedes Benz R&D India)

Bengaluru, India

Senior Engineer

Oct 2022 - May 2023

- Led an 8-member data-science team, analyzing Mercedes Benz e-bus powertrain operating in 5 German cities.
- Initiated Agile methodologies as Scrum master reducing analytics reports delivery time by 35%.
- Conducted research on Battery Lifetime Prediction of electric buses using data-driven methods.

Bosch Limited

Bengaluru, India

Assistant Manager

July 2019 - Oct 2022

- Led e-powertrain commissioning for 12 vehicles by diagnosing software issues utilizing CAN tools like CANalyzer.
- Published "A Study of Parameters Influencing Energy Consumption of an Electric Vehicle", IEEE ITEC-INDIA, 2019.
- Developed an electric powertrain simulation platform using Simulink and Python for efficient powertrain sizing.
- Conducted V&V for EV powertrain systems, including performance evaluation on chassis dynamometers.
- Generated drive cycles from test data using genetic algorithms to standardize controlled tests for EVs.

Intern

Jan 2019 - May 2019

- Innovated a precise 'Accelerator Pedal Calibration' method with Fuzzy Logic cutting calibration time by 20 hours.

Projects

Autonomous Highway Driving Using Deep Reinforcement Learning

- Developed an autonomous driving agent for the Highway environment, focusing on safe lane changing.
- Utilized Python with PyTorch and MuJoCo to train and evaluate the agent.
- Developed and evaluated variations of DQN, PPO, and DDPG algorithms to understand their strengths and trade-offs.

Autonomous Vehicle SLAM, Planning and Control

- Executed localization, path planning, and mapping for autonomous vehicles in a Unity simulation using ROS.
- Integrated Adaptive Monte Carlo Localization for precise vehicle localization within the mapped environment.
- Applied Hector SLAM for live mapping, enabling real-time environment mapping during navigation.
- Integrated RRT-based methods for planning and Pure Pursuit control for precise trajectory tracking.
- Created ROS-based Stereo Visual Odometry module for precise camera pose estimation using stereo images.

Road Segmentation Using VGG-FCN8

- Developed a road segmentation model using VGG-FCN8, adapting VGGNet for autonomous vehicle perception tasks.
- Designed a fully convolutional network based on the VGG16 architecture, adapting it for segmentation tasks by adding upsampling layers and incorporating skip connections.

Object Detection Using Vision Transformer

- Designed a novel detection model using vision transformers, feature pyramid networks and a YOLO detection head.
- Optimized localization accuracy using upsampling layers, skip connections, and customized loss functions.

Education

University at Buffalo, The State University of New York

Buffalo, NY

Master of Science in Robotics, CGPA 3.86

Aug 2023 - Dec 2024

Manipal Institute of Technology

Manipal, India

Bachelor of Technology in Mechanical Engineering

Aug 2015 - May 2019